Recitation 3 Solutions

Problem 1

Consider the "employee_survey_data.csv" dataset available on Courseworks.

The data was collected by the HR department of a mid-sized tech company with around 1,000 employees. The survey was part of an internal initiative to understand employee satisfaction, retention, and performance across various departments. It was intended to help HR and management identify factors that might influence turnover rates and develop targeted retention strategies.

HR used this dataset to investigate relationships between factors such as salary, job satisfaction, work hours, and turnover risk. The goal was to find patterns that could guide future decisions about work policies, benefits, and employee development programs. The dataset consists of 700 observations across 10 variables, with both numerical and categorical features. The features are:

- Employee_ID (Numeric) - Unique identifier for each employee
- Age (Numeric) - Age of employees (in years)
- Sex (Categorical) - Sex of the employee [Levels: Female, Male]
- Department (Categorical) - Department where the employee works [Levels: Sales, Engineering, HR, Marketing, Finance]
- Years_at_Company (Numeric) - Number of years the employee has been with the company
- Salary (Numeric) - Annual salary of the employee (USD)
- Job_Satisfaction (Categorical) - Employee's self-reported job satisfaction [Levels: Low, Medium, High]
- Work_Hours (Numeric) - Average number of hours worked per week
- Education_Level (Categorical) - Highest level of education attained by the employee [Levels: High School, Bachelor's, Master's, PhD]
- Turnout_Risk (Categorical) - Whether the employee is at risk of leaving the company [Levels: Yes, No]

```
employees <- read.csv("employee_survey_data.csv", stringsAsFactors = TRUE)
```

Question 1:

- How many employees are `Female`, have a `Master's` degree, and earn a `Salary` above $80,000?

```
nrow(employees[employees$Sex == "Female" & employees$Education_Level == "Master's" &

          employees$Salary > 80000, ])
```

```
## [1] 0
```

- What is the median `Work_Hours` for employees in the `Marketing` department versus the `Sales` department?

```
median_workhours <- tapply(employees$Work_Hours, employees$Department, median)

median_workhours["Marketing"]
```

```
## Marketing
##        45
```

```
median_workhours["Sales"]
```

```
## Sales
##    44
```

- What proportion of employees with `High` job satisfaction are at risk of turnover?

```
high_js <- employees[employees$Job_Satisfaction == "High", ]

mean(high_js$Turnover_Risk == "Yes")
```

```
## [1] 0.4920635
```

- Identify the department with the largest proportion of employees at risk of turnover.

```
turnover <- ifelse(employees$Turnover_Risk == "Yes", 1, 0)

dept_prop <- tapply(turnover, employees$Department, mean)

names(which.max(dept_prop))
```

```
## [1] "Engineering"
```

- For each `department`, calculate the ratio of `High job satisfaction` to `Low job satisfaction` employees.

```
high_count <- tapply(employees$Job_Satisfaction == "High", employees$Department, sum)

low_count <- tapply(employees$Job_Satisfaction == "Low", employees$Department, sum)

ratios <- high_count / low_count

print(ratios)
```

```
## Engineering     Finance          HR   Marketing       Sales
##   1.5135135   1.2272727   0.9333333   1.1395349   0.9807692
```

Question 2:

- Find the top 5 highest paid employees in each department. What is the average `Age` of these top earners?

```
depts <- unique(employees$Department)

top5_all <- data.frame()

for (d in depts) {

  dept_data <- employees[employees$Department == d, ]

  dept_sorted <- dept_data[order(dept_data$Salary, decreasing = TRUE), ]

  top5 <- head(dept_sorted, 5)

  top5_all <- rbind(top5_all, top5)
}


mean(top5_all$Age)
```

```
## [1] 41.2
```

- Compare the average `Work_Hours` of employees with `Salary` above the 75th percentile versus below the 25th percentile.

```
q25 <- quantile(employees$Salary, 0.25)

q75 <- quantile(employees$Salary, 0.75)


low_group  <- employees[employees$Salary < q25, ]

high_group <- employees[employees$Salary > q75, ]


mean_low  <- mean(low_group$Work_Hours)

mean_high <- mean(high_group$Work_Hours)


c("Below 25th" = mean_low, "Above 75th" = mean_high)
```

```
## Below 25th Above 75th
##   45.26857   45.35429
```

- Compute the variance and standard deviation of `Salary` for employees with ≤ 5 years versus > 10 years at the company.

```r
group1 <- employees[employees$Years_at_Company <= 5, "Salary"]

group2 <- employees[employees$Years_at_Company > 10, "Salary"]


results <- data.frame(

  Group = c("<= 5 years", "> 10 years"),

  Variance = c(var(group1), var(group2)),

  StdDev   = c(sd(group1), sd(group2)))


print(results)
```

```
##        Group   Variance   StdDev
## 1 <= 5 years 1299548258 36049.25
## 2 > 10 years 1152605489 33950.04
```

- Compute the number of employees with `Low` job satisfaction and `Salary` below the median salary.

```r
med <- median(employees$Salary)

low_med <- employees[employees$Job_Satisfaction == "Low" & employees$Salary < med, ]

nrow(low_med)
```

```
## [1] 112
```

Question 3:

- Add a new factor variable named `tenure_groups` with levels `Short`, `Medium`, and `Long`. Generate this variable based on the `Years_at_Company` variable without modifying the original `Years_at_Company` variable. The grouping is done as follows: employees with 0-5 years of tenure belong to the `Short` group, employees with 6-15 years of tenure belong to the `Medium` group, and employees with 16+ years belong to the `Long` group.

```r
yrs <- employees$Years_at_Company

groups <- ifelse(yrs <= 5, "Short", ifelse(yrs <= 15, "Medium", "Long"))

employees$tenure_groups <- factor(groups, levels = c("Short", "Medium", "Long"))
```

- Create a barplot to showcase the distribution of the `tenure_groups` variable. Properly label the plot and axes, change the colors of bars and borders. Also, change the bar names to "SHORT", "MEDIUM", and "LONG".

```r
barplot(
table(employees$tenure_groups),

col = c("lightsteelblue1", "wheat1", "thistle2"),

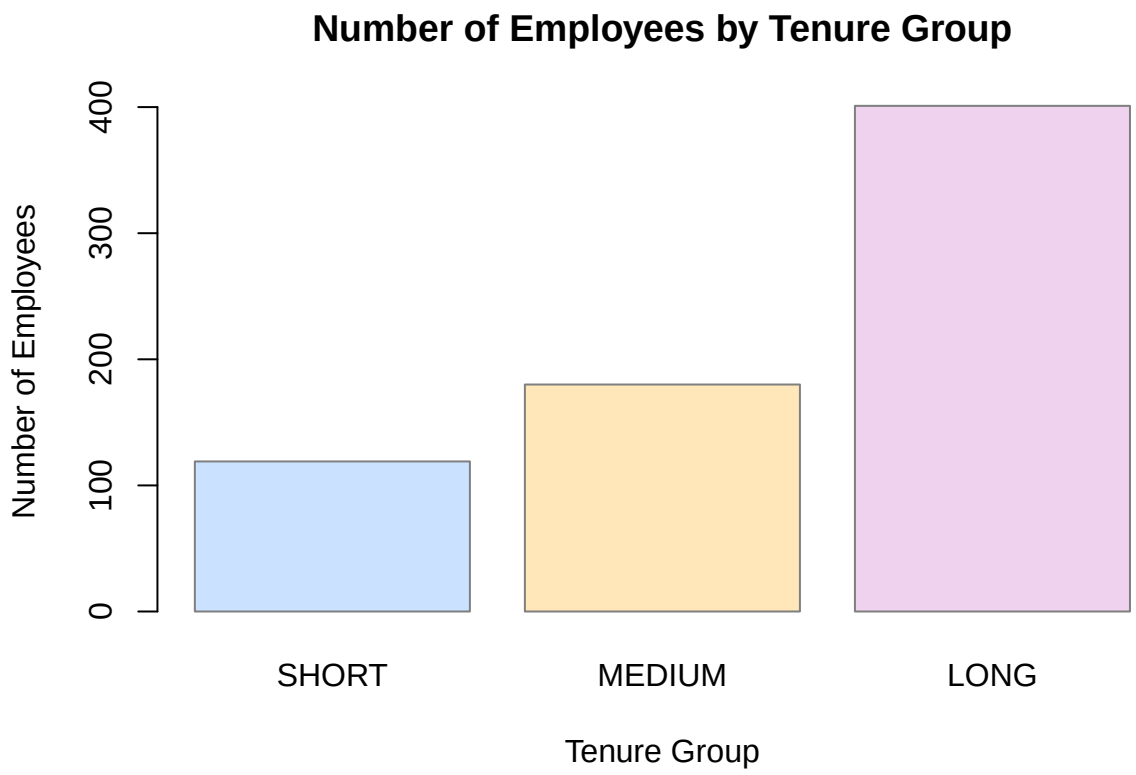
border = "grey50",

xlab = "Tenure Group",

ylab = "Number of Employees",

names.arg = c("SHORT", "MEDIUM", "LONG"),

main = "Number of Employees by Tenure Group"
)
```

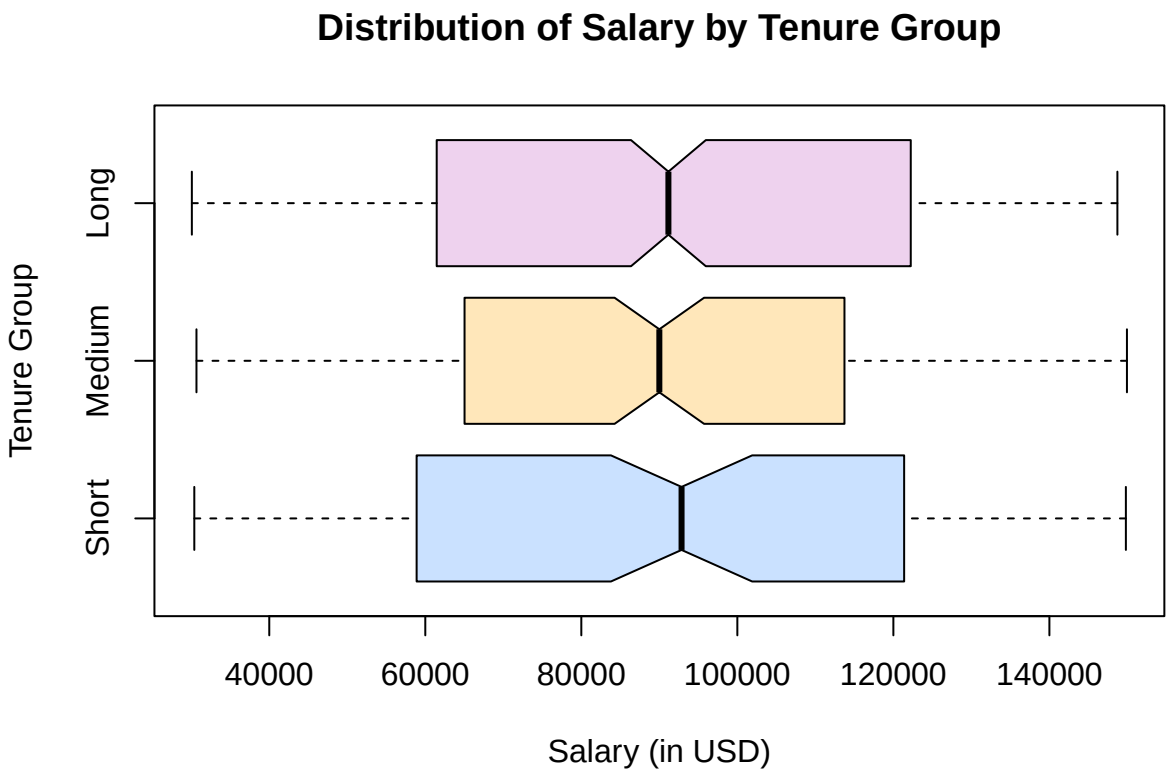


- Create a boxplot illustrating the distribution of `Salary` across different categories of the `tenure_groups` variable. Label the plot and axes appropriately, add a notch to each boxplot. Are there any outliers?

```r
boxplot(employees$Salary ~ employees$tenure_groups, horizontal = TRUE, notch = TRUE,

col = c("lightsteelblue1", "wheat1", "thistle2"), xlab = "Salary (in USD)",

ylab = "Tenure Group", main = "Distribution of Salary by Tenure Group")
```



There are no outliers.

Question 4:

- Compare the mean `Salary` values across various levels of `Job_Satisfaction`. Subsequently, eliminate all observations where `Work_Hours` exceeds 50 hours per week and repeat the preceding analysis. Elaborate on the impact of this filtering on the results.

```
aggregate(Salary ~ Job_Satisfaction, data = employees, FUN = mean)
```

```
##   Job_Satisfaction   Salary
## 1             High 90316.90
## 2              Low 89527.42
## 3           Medium 91402.17
```

```
doesnt_exceed_50hrs <- employees$Work_Hours <= 50

filtered_employees <- employees[doesnt_exceed_50hrs, ]

aggregate(Salary ~ Job_Satisfaction, data = filtered_employees, FUN = mean)
```

```
##   Job_Satisfaction   Salary
## 1             High 92815.84
## 2              Low 89829.69
## 3           Medium 90482.96
```

The average salary of employees with high or low levels of job satisfaction tends to be slightly higher than the overall average among those working at most 50 hours per week. However, for employees with medium levels of job satisfaction who do not exceed 50 hours per week of work, the average salary is slightly lower.

- Plot a histogram of the `Salary` feature before and after eliminating all observations where `Work_Hours` exceeds 50 hours per week, and display both histograms side by side. Analyze whether the shape of the distributions differ significantly after filtering. Customize the histograms by adding appropriate labels and a title, changing the colors of the bins, and setting the number of bins in each histogram to 10.

```r
par(mfrow = c(1, 2))

hist(
employees$Salary,

breaks = 10,

col = "lightsteelblue1",

xlab = "Salary (in USD)",

ylab = "Number of Employees",

main = "All Employees",

ylim = c(0, 70)
)

hist(
filtered_employees$Salary,

breaks = 10,

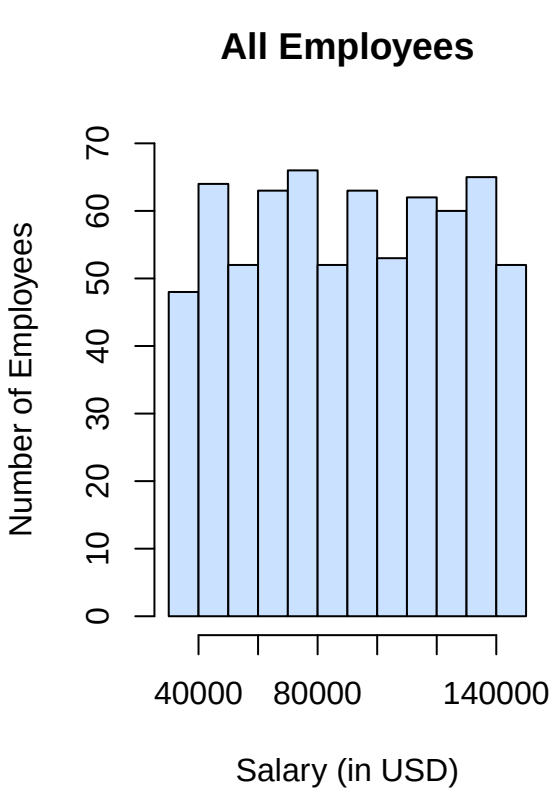
col = "lightsteelblue1",

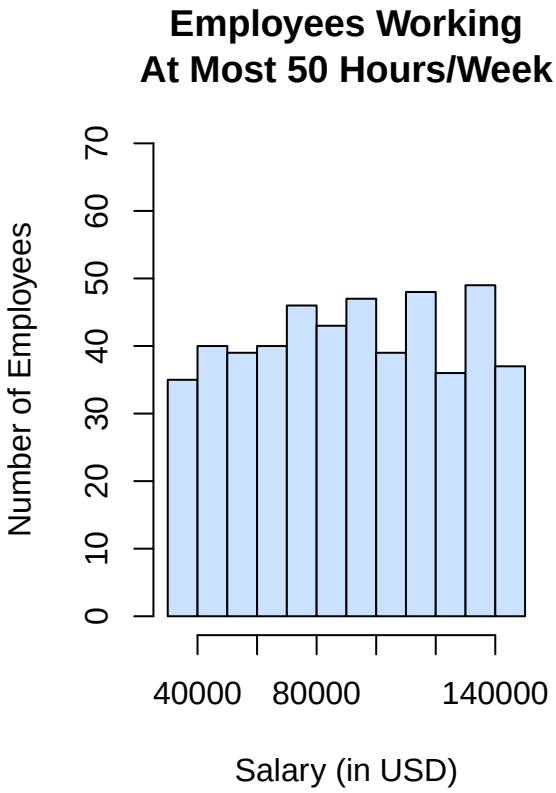
xlab = "Salary (in USD)",

ylab = "Number of Employees",

main = "Employees Working\nAt Most 50 Hours/Week",

ylim = c(0, 70)
)
```





The shapes of the distributions don't seem to differ significantly before and after filtering.